THE DATA ENGINEER LAB

DATABRICKS FROM ZERO TO HERO

A 30-Day Interview & Career Readiness Series

DAY 2

Databricks Architecture — Control Plane vs Compute Plane

SECTION 01

The Two Planes, At a Glance

Databricks' architecture is split into two logical planes: the Control Plane and the Compute Plane. Understanding this split is one of the most frequently tested architecture concepts in Databricks interviews.

- Control Plane: backend services Databricks manages in its own cloud account, including the web application, notebook interface, and job scheduling metadata.
- Compute Plane: where your actual data processing happens. For classic compute, this runs inside your own cloud account (AWS, Azure, or GCP).

CORE CONCEPT

The Control Plane coordinates and manages. The Compute Plane is where Spark clusters actually run and process data. This separation is why Databricks does not store or process your data inside its own infrastructure by default.

SECTION 02

What Lives Where

Inside the Control Plane

- The web application and user interface.
- Notebook commands and revision history, encrypted at rest.
- Job scheduling and cluster management services (start, stop, scale).
- Unity Catalog's control-plane metadata services.

Inside the Compute Plane

- Cluster virtual machines, the actual Spark executors doing the work.
- Your data in transit and in use during processing.
- For classic compute, this plane lives entirely inside your own cloud subscription.

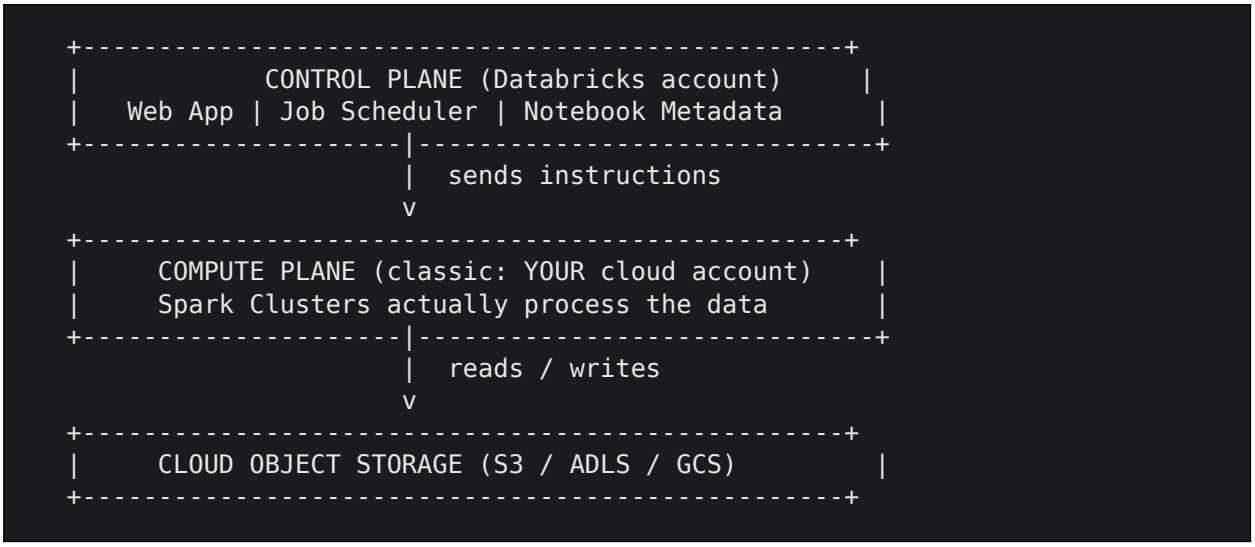
GOOD TO KNOW

Databricks also offers serverless compute, where the compute plane itself runs inside a Databricks-managed account rather than yours, trading some infrastructure control for convenience. Classic compute (the default for most notebooks and jobs) is the one that lives in your own cloud account.

SECTION 03

Architecture Diagram

A simplified view of how a job actually flows through both planes:



REMEMBER THIS FOR INTERVIEWS

A precise answer for this question: Databricks operates a Control Plane, managed by Databricks, and a Compute Plane, where data is actually processed. For classic compute this compute plane runs inside the customer's own cloud account, which is also why customer data stays within their own environment.

SECTION 04

Control Plane vs Compute Plane

Aspect	Control Plane	Compute Plane (Classic)
Managed By	Databricks	You (customer cloud account)
Contains	Web app, job scheduler, notebook metadata	Spark clusters, actual data processing
Where It Lives	Databricks' own cloud account	Your AWS, Azure, or GCP account
Security Relevance	Coordinates access and orchestration	Where your data actually stays and gets processed

SECTION 05

Glossary — Day 2 Terms

Term	Definition
Control Plane	Databricks-managed backend containing the web app, job scheduling, and notebook metadata.
Compute Plane	Where actual data processing occurs; classic compute runs this inside your own

	cloud account.
Classic Compute	The default compute model where cluster VMs run in the customer's own cloud subscription.
Serverless Compute	A model where the compute plane itself runs inside a Databricks-managed account.
Cluster	A set of virtual machines running Spark, used to execute your data processing jobs.

SECTION 06

Interview Questions & Answers

Q1. What are the two planes in Databricks' architecture?

A. The Control Plane, managed by Databricks, which contains the web application and job scheduling metadata, and the Compute Plane, where actual data processing happens.

Q2. Where does the Compute Plane run?

A. For classic compute, it runs inside the customer's own cloud account (AWS, Azure, or GCP). For serverless compute, it runs inside a Databricks-managed account instead.

Q3. Why does this split matter for security and compliance?

A. Because your actual data and processing stay inside your own cloud account for classic compute, giving you more control over data residency, network isolation, and compliance requirements.

Q4. Is notebook code stored in the Control Plane or Compute Plane?

A. Notebook commands and revision history are stored in the Control Plane, encrypted at rest, while the actual execution of that code happens in the Compute Plane.

Q5. What is the tradeoff between classic and serverless compute?

A. Classic compute gives you more infrastructure control since it runs in your own cloud account, while serverless compute trades some of that control for convenience, faster startup, and less infrastructure management.

STAY ON TRACK FOR ALL 30 DAYS

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Piyush Varma | The Data Engineer Lab